MTAT.07.003 Cryptology II
Spring 2012 / Exercise session ?? / Example Solution

Exercise (Polynomial message authentication code as almost universal hash function). *Consider messages that are ℓ-element lists $\boldsymbol{m} = (m_1, \ldots, m_\ell) \in \mathbb{F}^\ell$. Then the polynomial message authentication code is defined as follows*

$$h(\boldsymbol{m}, k_1, k_0) = m_\ell k_1^\ell + m_{\ell-1} k_1^{\ell-1} \cdots + m_1 k_1 + k_0 .$$

This code is not a universal hash function and thus there is an substitution attack that breaks the trivial success limit $\frac{1}{|\mathbb{F}|}$. The corresponding attack can be described as the following game

$$\begin{array}{l}\mathcal{G}_0\\ \left[\begin{array}{l} k_0, k_1 \xleftarrow{u} \mathbb{F}\\ \boldsymbol{m} \leftarrow \mathcal{A}\\ t \leftarrow h(\boldsymbol{m}, k_1, k_0)\\ \overline{\boldsymbol{m}}, \bar{t} \leftarrow \mathcal{A}(t)\\ \textbf{\textit{return}}\ [\boldsymbol{m} \neq \overline{\boldsymbol{m}}] \wedge [\bar{t} = h(\overline{\boldsymbol{m}}, k_1, k_0)] \end{array}\right.\end{array}$$

If h would be a universal hash function then the game would simplify to the following game

$$\begin{array}{l}\mathcal{G}_1\\ \left[\begin{array}{l} \boldsymbol{m} \leftarrow \mathcal{A}\\ t, t_* \xleftarrow{u} \mathbb{F}\\ \overline{\boldsymbol{m}}, \bar{t} \leftarrow \mathcal{A}(t)\\ \textbf{\textit{return}}\ [\boldsymbol{m} \neq \overline{\boldsymbol{m}}] \wedge [\bar{t} = t_*] \end{array}\right.\end{array}$$

for which the success is guaranteed to be at most $\frac{1}{|\mathbb{F}|}$. To estimate the statistical distance between the games $\mathcal{G}_0$ and $\mathcal{G}_1$ one has to estimate how much $h(\boldsymbol{m}, k_1, k_0), h(\overline{\boldsymbol{m}}, k_1, k_0)$ differ from $t, t_* \xleftarrow{u} \mathbb{F}$. Compute the corresponding distance and resulting upper bound on the attack success. Is the result the same as obtained by direct analysis of $\mathcal{G}_0$?

Solution. Hint: Just compute the probabilities.